

Skills Summary

Powers of i and the Complex Plane

Use this reference while completing your worksheet or when studying.

Skill 1 — The Four-Step Cycle of Powers of i

The cycle: $i^1 = i$ $i^2 = -1$ $i^3 = -i$ $i^4 = 1$

Rule: divide the exponent by 4 and keep the remainder. Remainder $0 \rightarrow 1$, $1 \rightarrow i$, $2 \rightarrow -1$, $3 \rightarrow -i$. A coefficient in front of the power just rides along.

Example: Simplify $2i^{19} + i^{24}$

Step 1. Both terms are powers of i with big exponents, so reduce each exponent modulo 4 before doing any arithmetic.

Step 2. For the first term the remainder is 3, so that power is $-i$ and the term becomes $-2i$. For the second the remainder is 0, so that power is 1.

$$\Rightarrow 1 - 2i$$

Try it: Simplify $3i^{22} - i^{13}$

check: $3 - 8 = -5$

Skill 2 — Points and Distances on the Plane

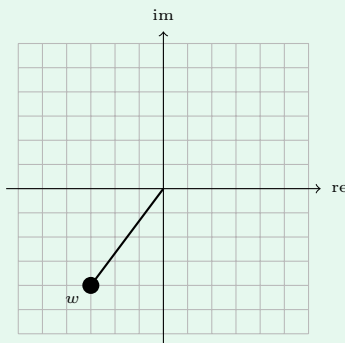
The picture: $a + bi$ is the point (a, b) — a along the horizontal *real* axis, b up the vertical *imaginary* axis.

Distance from the origin: the two parts are the legs of a right triangle, so the distance is $\sqrt{a^2 + b^2}$.

Example: Locate $w = -3 - 4i$ and find its distance from the origin.

Step 1. Negative real part means left of the origin and negative imaginary part means below it, so w sits in the lower-left quarter — then the two parts are the legs of a right triangle.

Step 2. The legs are 3 and 4, so the distance is the hypotenuse $\sqrt{9 + 16}$.



$$\Rightarrow 5$$

Try it: How far is $6 + 8i$ from the origin?

check: 10

Skill 3 — Multiplying by i Is a Quarter Turn

Rule: $i(a + bi) = -b + ai$. On the plane this is a quarter turn (90°) counterclockwise about the origin, and the distance from the origin does not change.

Two turns (i^2) is a half turn, which negates both parts; four turns (i^4) is all the way round, back to the start.

Example: Compute $i(4 - 3i)$

Step 1. Multiplying by i should turn the point a quarter turn, so expect the parts to swap and one sign to flip — distribute and check that against the answer.

Step 2. $4i - 3i^2$, and $-3i^2 = +3$, so the real part is 3 and the imaginary part is 4: the point $(4, -3)$ has turned to $(3, 4)$.

$$\Rightarrow \mathbf{3 + 4i}$$

Try it: Compute $i(-2 + 5i)$

check: $-5 - 2i$

(!) **Watch out:** the remainder 2 gives -1 , not $+1$ — the most common slip on this page is losing that sign in a term like $3i^{14}$. And a remainder of 0 means the power is 1, not 0.